Integrating Variational Modeling and Deep Learning for Removal of Mixed Complex Noise and Stripes in Hyperspectral Images

Dongyi Li, Huanfeng Shen, *Senior Member, IEEE*, Xiaobin Guan, *Member, IEEE*, Dekun Lin, Dong Chu, and Hongzhao Tang

***Abstract*—Hyperspectral imagery (HSI) is highly susceptible to severe contamination by mixed noise during acquisition and transmission, including Gaussian noise, impulse noise, and complex multidirectional (e.g., oblique or curvilinear) stripes. This article proposes an optimization-inspired deep network for complex noise removal in HSI (OIC-Net) that integrates the mathematical interpretability of model-based optimization with the representational power of deep learning for HSI mixed noise removal. The proposed framework deeply unrolls an iterative physical solver into an end-to-end architecture, seamlessly unifying variational optimization with deep neural networks. In particular, OIC-Net constructs an interactively coupled, dual-branch subnetwork mechanism dedicated to abundance map restoration and stripe elimination, achieving progressive noise suppression through efficient latent variable interaction in a single forward pass. Tailored to complex real-world degradations, an omnidirectional convolutional module is designed to explicitly encode structural priors for the effective isolation of oblique and multidirectional stripes. Concurrently, the abundance restoration subnetwork employs an encoder-decoder architecture to guarantee highly accurate spatial structure reconstruction. The proposed framework possesses explicit mathematical interpretability while demonstrating superior cross-sensor generalization and computational efficiency, enabling robust noise isolation and high-fidelity detail restoration under severe real-world degradations.**

***Index Terms*—Deep unfolding network, denoising, destriping, hyperspectral image, oblique stripes, variational optimization.**

I. Introduction

HYPERSPECTRAL imagery (HSI) simultaneously acquires rich spatial and spectral information from observed targets [1], [2]. Owing to its superior spectral characterization capability, HSI has been extensively applied in a wide range of applications, including vegetation monitoring [3], anomaly detection [4], and environmental protection [5]. However, due to sensor instability, atmospheric interference, and transmission errors, HSI is inevitably contaminated by various types of mixed noise during acquisition and transmission processes, including Gaussian noise, impulse noise, stripe noise, and radiometric distortion [6], [7]. Furthermore, after orthorectification of satellite imagery, originally vertical stripe artifacts are often transformed into oblique structures, as illustrated in Fig. 1(d). In addition, uncrewed aerial vehicle (UAV)-based imaging systems are highly susceptible to wind disturbances and mosaicking errors, resulting in stripe degradations with complex multidirectional or even curvilinear distributions, as shown in Fig. 1(c). Particularly in low-SNR bands, such as water vapor absorption regions, these degradations frequently coexist and couple with each other, forming highly complex mixed-noise scenarios. These degradations severely deteriorate the quality of HSI and consequently hinder subsequent interpretation and analysis tasks. Therefore, effective HSI quality enhancement and denoising before downstream applications are of great significance [8], [9].

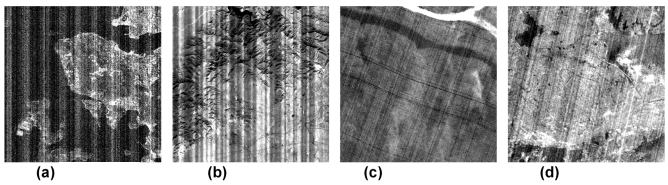


Fig. 1. Real HSI datasets heavily contaminated by various types of noise. (a) AHSI GF5. (b) OHS HIS. (c) UAV PanJin. (d) HJ1A HIS.

Current mainstream techniques for mixed noise removal in HSI are broadly categorized into optimization-model-based and deep learning-based approaches. By leveraging the intrinsic priors of clean HSI alongside the distinct structural properties of diverse degradations, optimization-based methods formulate unified mathematical frameworks driven by specific regularization terms. Prominent regularization strategies for characterizing HSI priors include low-rank (LR) matrix factorization to capture spectral redundancy [10], [11], [12], tensor decomposition (TD) to model global spatial-spectral correlations [13], [14], [15], [16], [17], [18], and total variation (TV) to enforce local spatial smoothness [19], [20]. For instance, the double-factor-regularized low-rank

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tensor factorization method (LRTF-DFR) [21] characterizes the global spectral LR property through tensor factorization, imposes weighted group sparsity on the spatial variations of the spatial factors, and enforces continuity on the spectral factors, thereby achieving favorable restoration performance. Despite the proliferation of algorithms driven by these properties, conventional optimization models frequently struggle to accurately decouple stripe noise from the underlying clean scenes. This limitation arises primarily because the spatial distribution of stripes is structurally analogous to high-frequency image gradients, such as inherent edges and textures. To address this structural ambiguity, existing studies have proposed specialized techniques explicitly tailored for stripe modeling. For instance, the combined global sparsity and local variation (CGSLV) method [22] formulates a prior that exploits the global sparsity and directional discontinuity of stripes. Similarly, the double low-rank (DLR) matrix factorization approach [23] mitigates stripe degradation by capitalizing on its spatial correlations. Nevertheless, owing to significant discrepancies in scene semantics, data distributions, and noise intensities across different HSIs, most of these model-driven techniques exhibit pronounced parameter sensitivity and limited generalization. Furthermore, the complex iterative nature of these numerical optimization algorithms inherently incurs prohibitive computational overhead and prolonged execution times, restricting their practical applications.

In recent years, deep learning-based architectures have been extensively deployed for HSI denoising. CNNs served as the pioneering models for HSI restoration, inspiring a wide array of CNN-derived architectures [24], [25]. However, conventional spatial convolutions inherently struggle to explicitly model the contiguous spectral correlations of HSI. To bridge this gap, RNNs and their sequential variants were introduced, leveraging their superiority in sequence modeling to effectively capture long-range spectral dependencies [26], [27]. More recently, to overcome the inherent limitations of convolutional kernels in modeling global contextual information, self-attention mechanisms have emerged as a dominant paradigm in HSI denoising [28], [29], [30]. For instance, the spatial-spectral Transformer (SST) [31] couples nonlocal spatial attention with global spectral attention within a unified framework, significantly augmenting the capacity to characterize long-range dependencies. The universal restoration framework (MP-HSIR) [32] that integrates spectral, textual, and visual prompts into a prompt-guided spatial-spectral Transformer to address diverse degradation types and intensities. Dixit et al. [33] integrated supervised contrastive learning and image inpainting to enhance feature learning for HSI denoising. Despite these representational advancements, the majority of these purely data-driven networks operate as uninterpretable "black boxes." By relying exclusively on massive data fitting, they severely neglect the intrinsic physical generation mechanisms and structural priors of HSI [34], [35]. Consequently, lacking explicit mathematical constraints, these models frequently suffer from limited generalization, struggling to maintain robust denoising fidelity when confronting complex real-world mixed-noise scenarios.

Despite the development of numerous HSI denoising methods, effectively addressing the mixed degradation induced by diverse complex noises remains a formidable challenge. Conventional optimization-based approaches typically formulate this task as an illposed inverse problem [36], [37], relying on hand-crafted regularizers derived from idealized structural priors of the noise. However, in real-world scenarios, structural degradations often manifest as highly intricate, multidirectional, or even curvilinear patterns [e.g., Fig. 1(c) and (d)]. This structural complexity is severely amplified in a low-SNR channels, fundamentally violating the underlying assumptions of conventional prior-driven designs. Consequently, rigidly formulated physics-based regularizers struggle to accurately characterize the superposition of heterogeneous mixed noises, inevitably introducing processing artifacts and spectral distortions [38], [39], [40]. Conversely, data-driven deep learning methods predominantly rely on opaque neural architectures governed by massive, indeterminate parameters, resulting in inherently uninterpretable models. Furthermore, by operating as entirely "black-box" systems, these approaches typically fail to explicitly embed domain-specific prior knowledge of the physical degradation mechanisms. Consequently, they often lack the structural sensitivity required to decouple and eliminate pronounced stripe artifacts. When confronting real-world scenarios characterized by complex, mixed-noise distributions, the performance of these purely data-driven models frequently deteriorates or entirely collapses.

To synergistically integrate the mathematical interpretability of model-based optimization with the representational power of deep learning, various hybrid approaches have recently emerged for HSI denoising [41], [42], [43]. Li et al. [44] proposed a robust deep recovery model that combines a pretraining-free deep image prior with an SSTGV regularizer and truncated-Cauchy data fidelity to jointly exploit spatial–spectral correlations and suppress mixed noise. Despite their usage of physical priors to guide network design, these methods still exhibit critical limitations in complex real-world scenarios. On the one hand, the direct migration of generic neural architectures fails to embed explicit physical encodings. Consequently, these models remain heavily reliant on massive training datasets, suffering from severe generalization degradation under varying sensor distributions and hardware aging. On the other hand, current hybrid frameworks struggle to decouple intricate, multidirectional structural degradations—such as oblique stripes induced by UAV attitude jitter or satellite orthorectification—which inherently violate the strict sparsity assumptions of traditional hand-crafted regularizers. Motivated by these fundamental bottlenecks, we propose an optimization-inspired deep network for complex noise removal in HSI (OIC-Net), which deeply unrolls an iterative physical solver into an end-to-end architecture. By constructing explicit physics-encoded neural layers, OIC-Net seamlessly unifies variational optimization with deep learning, thereby significantly bolstering mathematical interpretability and cross-sensor generalization. Furthermore, meticulously tailored to multidirectional stripe characteristics and intrinsic spatial–spectral priors, a specialized dual-branch architecture is devised to efficiently decouple complex real-world

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degradations, ensuring robust elimination of oblique stripes while maintaining strong inference stability. The main contributions of this article are summarized as follows.

1) A physics-driven deep unfolding framework for HSI mixed noise removal. By encapsulating an interpretable physical solver as the foundational operator, our model perfectly integrates explicit mathematical constraints with robust data-driven feature learning. This paradigm exhibits a superior generalization, band-agnostic flexibility, and high computational efficiency, enabling faithful detail restoration under severe real-world degradations.
2) An interactively coupled dual-subnetwork mechanism. Guided by the unfolded optimization steps, we devise two specialized modules for abundance map restoration and stripe elimination. Latent variables across these modules interact efficiently, fostering a mutually reinforcing mechanism that achieves iterative noise suppression and progressive detail recovery seamlessly within a single forward pass.
3) Tailored structural designs for oblique stripes and topological reconstruction. We propose an omnidirectional convolutional module to explicitly encode the directional priors of real-world oblique stripes, efficiently isolating complex structural artifacts. Concurrently, the abundance restoration subnetwork employs an encoder–decoder architecture, leveraging clean, low-resolution spatial information as guidance to guarantee highly accurate structural reconstruction.

II. Methods

A. Variational Model Formulation

Assume $\mathcal{Y}$ denotes the observed HSI contaminated by hybrid noise. Its degradation process can be formulated as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{B} \tag{1}$$

where $\mathcal{X} \in \mathbb{R}^{M\times N\times B}$ denotes the clean HSI, and $\mathcal{B} \in \mathbb{R}^{M\times N\times B}$ represents the stripe noise. Given that the stripe noise exhibits pronounced spatial structural characteristics distinct from random noise, we explicitly model it as an independent component. In general, (1) defines an illposed inverse problem that aims to separate the latent clean HSI $\mathcal{X}$ from the noisy observation $\mathcal{Y}$.

The severity of pixelwise noise (e.g., Gaussian and impulse noise) varies across different sensors and HSI bands, which prompts us to adopt a data-driven network for adaptive pixelwise noise removal. Conversely, structural noise (e.g., deadlines and stripes) possesses directional extension and complex structures (see Fig. 1). This motivates us to model it separately and devise a dedicated network that accounts for its structural features. In particular, we cast the HSI restoration as an optimization problem and detail the aforementioned terms

$$\{\hat{\mathcal{X}}, \hat{\mathcal{B}}\} = \arg\min_{\hat{\mathcal{X}},\hat{\mathcal{B}}} \left\|\mathcal{Y} - \hat{\mathcal{X}} - \hat{\mathcal{B}}\right\|_F^2 + \eta\left(\hat{\mathcal{X}}\right) + \gamma\left(\hat{\mathcal{B}}\right). \tag{2}$$

Operating directly in the high-dimensional HSI space entails prohibitive computational costs and substantial memory overhead. Based on the linear spectral mixture model [45], [46], [47], [48], noise components within the HSI are inherited by the abundance fractions (see Fig. 2); crucially, these maps reside in a subspace with much lower dimensionality. This drives us to employ a physical degradation model to derive initial abundance maps, thereby shifting the restoration process to the abundance domain. The corresponding model is expressed as

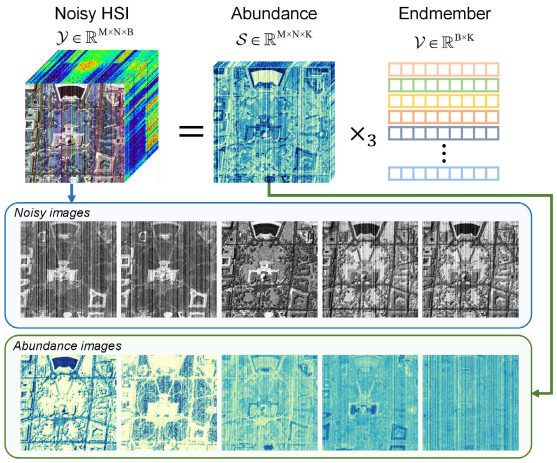


Fig. 2. Comparison between noisy HSI Image and abundance image.

$$\begin{aligned}\{\hat{\mathcal{S}}, \hat{\mathcal{B}}\} &= \arg\min_{\hat{\mathcal{S}},\hat{\mathcal{B}}} \left\|\mathcal{S}_{obs} - \hat{\mathcal{S}} - \hat{\mathcal{B}}\right\|_F^2 + \eta\left(\hat{\mathcal{S}}\right) + \gamma\left(\hat{\mathcal{B}}\right) \\ &\text{s.t. } \mathcal{X} = \mathcal{S} \times_3 \mathcal{A}\end{aligned} \tag{3}$$

where $\hat{\mathcal{S}}$ represents the potential clean abundance image, $\mathcal{S}_{obs}$ denotes the noisy abundance representation derived from the observed HSI, $\hat{\mathcal{B}}$ is the stripe noise present in the abundance image. The endmember matrix A is initialized using VCA on the observed HSI. Given A, the initial abundance maps are estimated by fully constrained least-squares unmixing under the abundance nonnegativity and sum-to-one constraints. To solve the nonconvex optimization problem in (3), variable splitting techniques are commonly used [49]. By introducing auxiliary variables, the original problem is decomposed into several subproblems that are easier to solve. With the help of the variable splitting technique, we introduce two auxiliary variables $\mathcal{H}, \mathcal{W}$ and reformulate the constrained optimization problem in (3). The formulation can be defined as

$$\begin{aligned}\{\hat{\mathcal{S}}, \hat{\mathcal{B}}\} &= \arg\min_{\hat{\mathcal{S}},\hat{\mathcal{B}}} \left\|\mathcal{S}_{obs} - \hat{\mathcal{S}} - \hat{\mathcal{B}}\right\|_F^2 + \eta\left(\mathcal{H}\right) + \gamma\left(\mathcal{W}\right) \\ &\text{s.t. } \mathcal{H} = \mathcal{S}, \mathcal{W} = \mathcal{B}.\end{aligned} \tag{4}$$

According to the half-quadratic splitting method, the cost function is then transformed into

$$\begin{aligned}L\left(\hat{\mathcal{S}}, \mathcal{H}, \mathcal{B}, \mathcal{W}\right) = &\left\|\mathcal{S}_{obs} - \hat{\mathcal{S}} - \mathcal{B}\right\|_F^2 + \eta\left(\mathcal{H}\right) + \gamma\left(\mathcal{W}\right) \\ &+ \mu\left\|\mathcal{H} - \hat{\mathcal{S}}\right\|_F^2 + \lambda\left\|\mathcal{W} - \mathcal{B}\right\|_F^2\end{aligned} \tag{5}$$

where μ, λ are the parameters that control the consistency between the auxiliary variables and the original variables.

B. Coupled Variational Optimization and Deep Learning Framework

In conventional optimization paradigms, solving (5) typically entails an iterative process of updating the variables

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by imposing manually crafted physical regularizers, such as $\eta(\mathcal{H})$ and $\gamma(\mathcal{W})$. However, rigidly defined mathematical priors inherently struggle to characterize the severe nonlinearity and complex distributions of real-world mixed noise. Furthermore, the iterative numerical updates inevitably suffer from prohibitive computational overhead and slow convergence. To circumvent these fundamental bottlenecks, we deeply unroll the iterative HQS optimization process into an end-to-end neural network architecture. In particular, instead of relying on the explicit physical regularizers, we design sophisticated learning-driven subnetworks to act as the proximal operators. These subnetworks implicitly model the latent variables $\mathcal{H}$ and $\mathcal{W}$ in a data-driven manner. By encapsulating the optimization algorithm into network layers, our proposed method successfully couples the mathematical interpretability of HQS with the powerful nonlinear representation capabilities of deep learning, thereby achieving robust and efficient complex noise removal.

In particular, the forward propagation process within each unfolded optimization layer can be mathematically formulated as

$$\begin{cases} \hat{\mathcal{S}}^{k+1} = \arg\min_{\hat{\mathcal{S}}} \left\| \mathcal{S}_{obs} - \hat{\mathcal{S}} - \mathcal{B}^k \right\|_F^2 + \mu \left\| \mathcal{H}^k - \hat{\mathcal{S}} \right\|_F^2 \\ \hat{\mathcal{H}}^{k+1} = \arg\min_{\hat{\mathcal{H}}} \mu \left\| \mathcal{H} - \hat{\mathcal{S}}^{k+1} \right\|_F^2 + \eta(\mathcal{H}) \\ \hat{\mathcal{B}}^{k+1} = \arg\min_{\hat{\mathcal{B}}} \left\| \mathcal{S}_{obs} - \hat{\mathcal{S}}^{k+1} - \mathcal{B} \right\|_F^2 + \lambda \left\| \mathcal{W}^k - \hat{\mathcal{B}} \right\|_F^2 \\ \hat{\mathcal{W}}^{k+1} = \arg\min_{\hat{\mathcal{W}}} \lambda \left\| \mathcal{W} - \hat{\mathcal{B}}^{k+1} \right\|_F^2 + \gamma(\mathcal{W}). \end{cases} \tag{6}$$

Each unfolded stage corresponds to one iteration of the HQS-based alternating optimization algorithm and contains a two learned proximal mappings and two explicit data-consistency updates.

For the $\hat{\mathcal{S}}$-subproblem, the clean abundance estimate is updated through an explicit gradient-descent layer

$$\begin{aligned} \hat{\mathcal{S}}^{k+1} &= \hat{\mathcal{S}}^k - \varepsilon_k \frac{\partial \mathcal{L}}{\partial \hat{\mathcal{S}}^{k+1}} \\ &= \hat{\mathcal{S}}^k - \varepsilon_k \left[\hat{\mathcal{S}}^k + \mathcal{B}^k - \mathcal{S}_{\text{obs}} + \mu_k \left(\hat{\mathcal{S}}^k - \mathcal{H}^k \right) \right]. \end{aligned} \tag{7}$$

For the $\mathcal{H}$-subproblem, the proximal mapping associated with the clean-abundance regularizer $\eta(\mathcal{H})$ is approximated by neural network $\eta_\theta(\cdot)$

$$\mathcal{H}^{k+1} = prox\left(\hat{\mathcal{S}}^{k+1}\right) \approx \eta_\theta^k\left(\hat{\mathcal{S}}^{k+1}\right). \tag{8}$$

The stripe component is subsequently updated using another explicit gradient-descent layer

$$\begin{aligned} \hat{\mathcal{B}}^{k+1} &= \hat{\mathcal{B}}^k - \varepsilon_k \frac{\partial \mathcal{L}}{\partial \hat{\mathcal{B}}^{k+1}} \\ &= \hat{\mathcal{B}}^k - \varepsilon_k \left[\hat{\mathcal{S}}^{k+1} + \hat{\mathcal{B}}^k - \mathcal{S}_{\text{obs}} + \lambda_k \left(\hat{\mathcal{B}}^k - \mathcal{W}^k \right) \right]. \end{aligned} \tag{9}$$

For the $\mathcal{W}$-subproblem, the proximal mapping associated with the stripe regularizer $\gamma(\mathcal{W})$ is approximated by neural network $\gamma_\theta(\cdot)$

$$\mathcal{W}^{k+1} = prox\left(\hat{\mathcal{B}}^{k+1}\right) \approx \gamma_\theta^k\left(\hat{\mathcal{B}}^{k+1}\right). \tag{10}$$

The two GDR modules in Fig. 3 directly implement (7) and (9). The abundance GDR combines the observation-fidelity gradient with the Multiscale Dense Residual Network (MDRN)-guided consensus term to update $\hat{\mathcal{S}}$, while the stripe GDR combines the fidelity gradient with the Stripe Pattern Detection and Removal Network (SPDN)-guided consensus term to update $\hat{\mathcal{B}}$. Both updates are scaled by their corresponding learnable step sizes. The updated pair $(\hat{\mathcal{S}}^{k+1}, \hat{\mathcal{B}}^{k+1})$ is passed to the next unfolded stage. Consequently, $\eta_\theta(\bullet)$ and $\gamma_\theta(\bullet)$ implement the two learned proximal operators, whereas the updates of $\hat{\mathcal{S}}$ and $\hat{\mathcal{B}}$ explicitly preserve the observation-model consistency. Cascading K such stages is equivalent to unfolding K iterations of the original HQS solver. The penalty parameters are implemented as stage-dependent learnable scalars rather than fixed hyperparameters shared across all stages. In particular, μ_k controls the consistency between $\hat{\mathcal{S}}^k$ and $\mathcal{H}^k$, while λ_k controls that between $\hat{\mathcal{B}}^k$ and $\mathcal{W}^k$. Their positivity is ensured by

$$\mu_k = softplus\left(\tilde{\mu}_k\right) + \delta, \lambda_k = softplus\left(\tilde{\lambda}_k\right) + \delta \tag{11}$$

where the unconstrained variables are jointly optimized with the network weights through backpropagation. The learned parameters remain fixed during inference, enabling the different stages to adaptively balance data consistency and learned prior constraints.

Through the formulation of (7)–(11), we construct a complete unfolded optimization stage. By cascading K such identical stages (the impact of K on restoration performance is comprehensively analyzed in Section III), we establish a deep iterative forward-propagation architecture. As the degraded input traverses these cascaded layers, the complex noise components are progressively decoupled and stripped away from the original abundance map, ultimately yielding the latent clean reconstruction. The fundamental superiority of this workflow lies in the profound mechanism by which it couples variational optimization with deep learning. Rather than treating the neural network as an uninterpretable black box, we leverage the derived variational optimization algorithm to dictate the exact architectural design. In particular, the mathematical derivation "hard-codes" the explicit physical constraints and mutual variable interactions directly into the network's forward pass. Concurrently, the tailored deep learning modules act as proximal operators, utilizing data-driven learning to implicitly extract the highly nonlinear priors of complex mixed noise. This synergistic coupling guarantees that the network adheres to strict physical principles while maintaining powerful representation capabilities, thereby significantly enhancing the training stability and robust generalization during inference. The overall workflow of the proposed framework is illustrated in Fig. 3.

C. Network Modules for Denoising and Destriping

This section details the proposed architecture, which comprises two core modules: the MDRN for pixel-level noise removal and the SPDN for structural stripe noise elimination.

1) Multiscale Dense Residual Reconstruction Module: To address the $\mathcal{H}$ subproblem, our objective is to remove pixel-level noise from the abundance images. We employ the MDRN to tackle this, with its specific architecture illustrated in Fig. 4. The formulated optimization problem can be expressed as

$$\mathcal{H}^{k+1} = \text{MDRN}\left(\hat{\mathcal{S}}^{k+1}\right). \tag{12}$$

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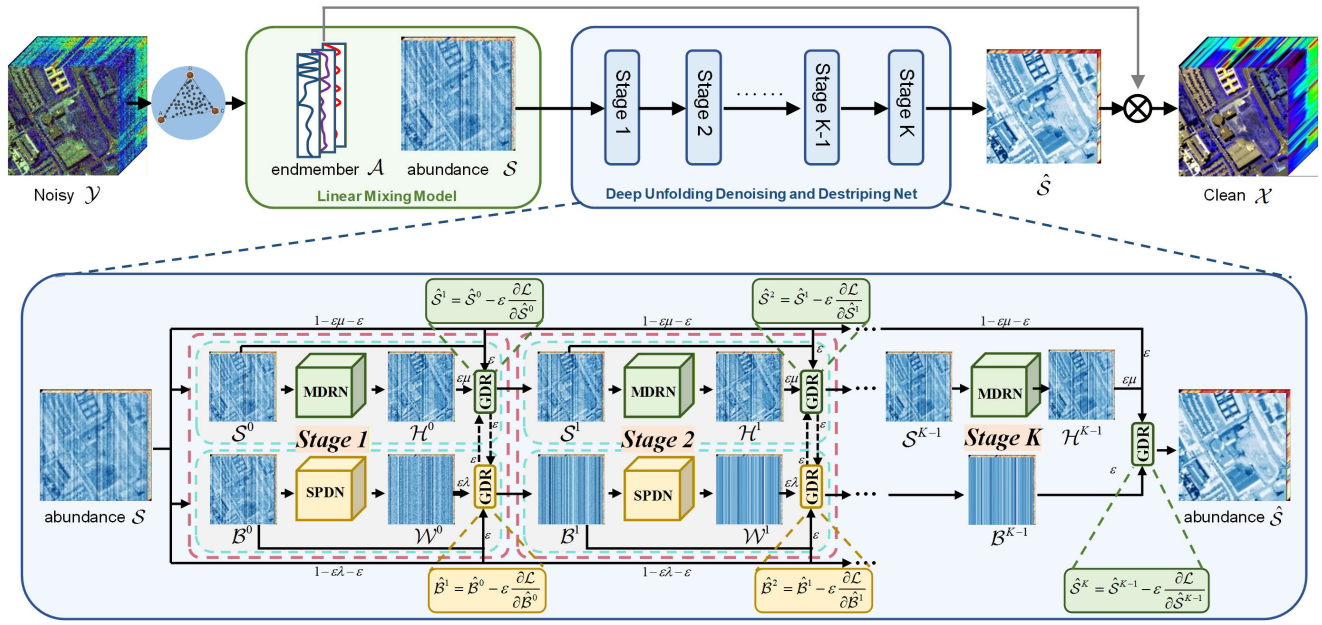


Fig. 3. Overall framework of OIC-Net.

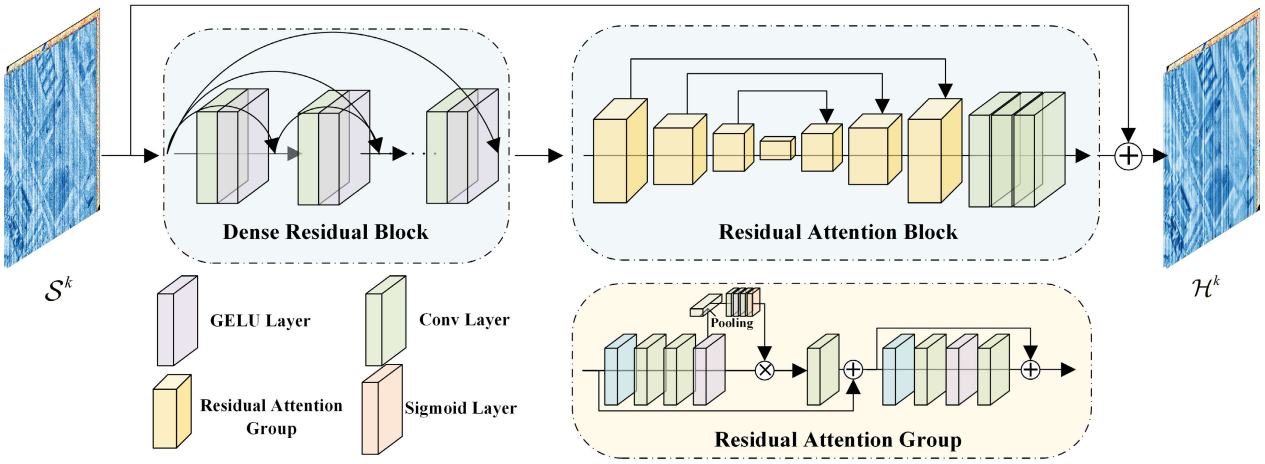


Fig. 4. Multiscale dense residual network.

The MDRN comprises a dense residual block (DRB) for initial feature encoding and a multiscale residual attention block (RAB) for hierarchical restoration. Utilizing the dense skip connections and GELU activations, the DRB efficiently extracts and fuses local–global features to ensure adaptive representation learning [43]. Subsequently, the RAB processes these preliminary features through a symmetric, four-stage encoder–decoder architecture. Within the encoder, residual attention groups (RAGs) integrate layer normalization (LN), local convolutions (3×3 and 1×1), and channel attention mechanisms. This configuration adaptively recalibrates informative channels while maintaining training stability and mitigating internal covariate shift. By employing a stride factor of 2, the downsampling process intrinsically suppresses high-frequency random noise, yielding a robust low-resolution representation. The symmetric decoder then utilizes transposed convolutions to iteratively restore spatial dimensions, assisted by skip connections that propagate high-resolution, low-level details from the encoder to guide deep restoration. Finally, a residual-based reconstruction layer generates the latent clean abundance image. Despite its efficacy in mitigating unstructured Gaussian and impulse noise, the MDRN struggles to eliminate conspicuous stripe artifacts. Because stripe noise typically exhibits strong directional and global structural characteristics, standard convolutions with localized receptive fields inherently fail to decouple it from the underlying image content. To overcome this fundamental limitation, we propose a dedicated destriping network meticulously designed for structural noise removal.

2) Stripe Detection and Removal Module: For the $\mathcal{W}$ subproblem, the objective is to eliminate stripe noise from the abundance images. To this end, we introduce the SPDN. The corresponding formulation is written as

$$\mathcal{W}^{k+1} = \mathrm{SPDN}\left(\mathcal{B}^{k+1}\right). \tag{13}$$

Unlike pixelwise noise distributions (e.g., Gaussian noise), stripe noise exhibits pronounced directional and structural characteristics. Traditional convolutions, while adept at handling pixel-level noise, are suboptimal for capturing these structural degradation patterns. By exploiting the directional

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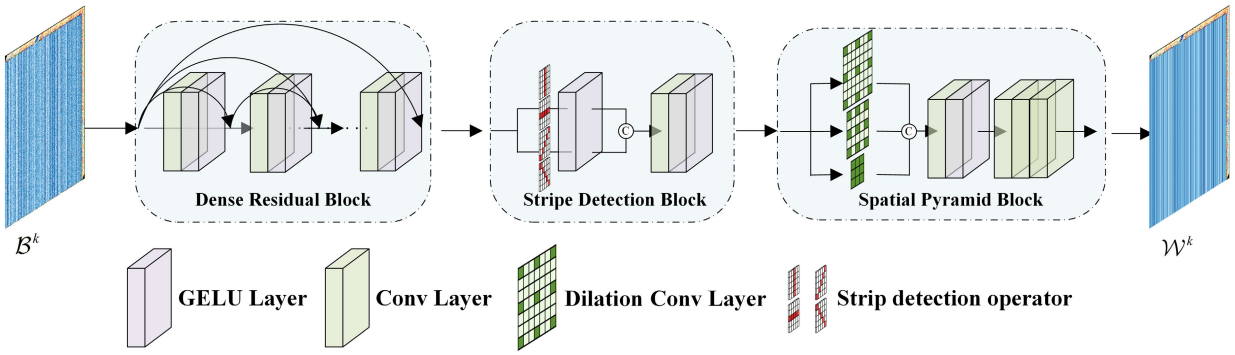


Fig. 5. Sparse pyramid striping removal network.

correlation of stripe noise in the gradient domain, we designed a learnable orientation-aware convolutional stripe detection module and a sparse pyramid convolutional stripe removal module. The learnable orientation-aware. The detection module disentangles multidirectional texture features using parameterized convolutional kernels oriented in the vertical, horizontal, and two orthogonal diagonal directions. This mechanism not only mitigates standard vertical and horizontal stripes but also effectively captures complex, multidirectional stripe patterns typical in UAV-acquired HSI data. Furthermore, the sparse pyramid removal module stacks multiple convolutional layers with varying dilation rates, expanding the receptive field to align with the sparse spatial distribution of stripe noise. These cascaded dilated convolutions form a multiscale receptive field, enabling the progressive extraction of stripe components via sparse sampling. In particular, as depicted in Fig. 5, preliminary features are extracted using a DRB. These features are subsequently fed into the learnable orientation-aware detection module, which utilizes 5×1 and 1×5 convolutional blocks to detect vertical and horizontal stripes, respectively. To handle oblique stripe noise caused by UAV platform jitter, the diagonal and antidiagonal convolutional paths are implemented by initializing 5×5 convolution kernels and filtering them with corresponding binary masking matrices. The features extracted from these four directional perception paths are aggregated through a 1×1 convolutional layer for preliminary fusion. A subsequent fusion layer, comprising a 1×1 convolution and a GELU activation, further integrates these multidirectional features. The processed data is then fed into the sparse pyramid convolutional stripe removal module. This module consists of three parallel 3×3 convolutions with dilation rates of 1, 2, and 3, respectively. This configuration expands their effective receptive fields to $3 \times 3, 5 \times 5$, and 7×7 without sacrificing spatial resolution. As the dilation rate increases, the weights of each convolutional layer become sparser [51], which facilitates the network's ability to model the sparse properties of stripe noise. Finally, the multiscale features from these three dilated paths are aggregated using a 1×1 convolution and a GELU activation to output the final destriped image.

D. Loss Function

To effectively preserve high-frequency details and robustly capture sparse noise distributions without the outlier sensitivity inherent to the L_2 norm, OIC-Net is trained end-to-end using the L_1 loss

$$\mathcal{L}_\theta = \left\| \mathcal{F}_\Theta(\mathcal{Y}) - \mathcal{X}_{gt} \right\|_1 \tag{14}$$

where Θ, $\mathcal{F}_\Theta(\mathrm{Y})$, and X denote the learnable parameters, the restored output, and the clean reference HSI, respectively. By explicitly integrating the spectral unmixing model, we constrain feature extraction within the abundance domain while computing the final loss in the HSI domain. This model-driven deep learning paradigm inherently provides a twofold advantage: it leverages data-driven modules to learn complex, nonlinear abundance priors that evade traditional analytical modeling, while utilizing embedded physical constraints to strictly regularize the network's solution space. Consequently, this deep integration significantly enhances both the interpretability and robust generalization of the framework.

III. Experiments

A. Experimental Setting

1) Datasets: To train the proposed OIC-Net for simulated experiments, we utilized the publicly available noise-free ICVL dataset. The original images ($1392 \times 1300 \times 31$) were randomly cropped into spatial-spectral cubes of $128 \times 128 \times 31$. To enhance model robustness, data augmentation techniques, including rotation and scaling, were applied, ultimately generating 20 000 training samples. Simulated testing was subsequently conducted on the Pavia ($340 \times 610 \times 80$) and WDC ($256 \times 256 \times 191$) datasets. In the real-data experiments, we selected three satellite datasets and one airborne dataset to validate the performance of OIC-Net.

1) *HJ1-A Dataset ($628 \times 596 \times 115$):* This dataset features a 100 m spatial resolution spanning the visible and near-infrared (VNIR) spectrum. Because the platform directly outputs georectified products, the structural degradation is predominantly manifested as oblique stripe noise, compounded by Gaussian noise, impulse noise, and dead lines.
2) *Inner Mongolia UAV Dataset ($3072 \times 1024 \times 335$):* Acquired by a UAV-mounted push-broom imager, this dataset provides a high spatial resolution of 1.92 m across a broad spectrum. Due to platform attitude instability (e.g., wind perturbations) and intrinsic sensor limitations, these images are severely corrupted by complex oblique stripes, radiation nonuniformity, and Gaussian noise.

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TABLE I
TRAINING CASES AT EACH STAGE

Epoch	Training settings
0~30	Gaussian Noise
31~60	Gaussian Noise+Impulse Noise
61~100	Gaussian Noise + Impulse Noise + Fixed Stripes
101~200	Gaussian Noise + Impulse Noise + Random Stripes
201~300	Gaussian Noise + Impulse Noise + Vertical Stripes
301~350	Gaussian Noise + Impulse Noise + Horizontal Stripes
351~400	Gaussian Noise + Impulse Noise + -90~0° Stripes
401~450	Gaussian Noise + Impulse Noise + 0~90° Stripes

3) *GF-5 Dataset (*$1000 \times 1000 \times 295$*):* Featuring a 30 m spatial resolution and a fine spectral resolution (~5 nm) covering 390–2504 nm. In this study, we specifically evaluated the heavily degraded short-wave infrared (SWIR) bands, which are severely afflicted by wide stripe artifacts induced by radiometric distortion, alongside mixed Gaussian and impulse noise.
4) *OHS Dataset (*$2000 \times 2000 \times 32$*):* Collected by the Zhuhai-1 push-broom imager with a 10 m spatial resolution covering 400–2500 nm. Affected by sensor limitations and atmospheric interference, this dataset is primarily contaminated by intense mixed Gaussian noise, radiometric distortion, and pronounced stripe artifacts.

2) Implementation and Model Training: The proposed OIC-Net was optimized using the Adam algorithm ($\beta_1 = 0.9$, $\beta_2 = 0.99$) with a batch size of 8. The learning rate was initialized at 10^{-3} and decayed to 10^{-6} via a piecewise schedule to ensure stable convergence. To effectively handle the complexity of mixed degradations, we employed a progressive multistage training strategy totaling 450 epochs. As detailed in Table I, each training phase progressively introduced a more challenging degradation configuration, with the network weights initialized from the preceding stage. Furthermore, to mitigate domain shifts caused by disparate sensor characteristics, the ICVL-pretrained model was fine-tuned using a limited number of reference-quality patches selected from additional scenes acquired by the corresponding sensors. In particular, regions containing visible stripes, dead lines, saturation, clouds, or severe atmospheric absorption were excluded, and the remaining high-SNR regions were cropped into 128×128 patches. Synthetic mixed noise was then added following the progressive training settings. These fine-tuning samples were spatially separated from the evaluated regions and were used only for limited sensor-specific adaptation. The overall training strategy is illustrated in Fig. 6. The deep learning models were trained on a Linux system equipped with an NVIDIA Tesla A100 GPU (40 GB). Conversely, all conventional model-driven benchmarks were executed using MATLAB R2021 on an AMD Ryzen 9 5950X CPU.

3) Compared Methods: To evaluate the proposed OIC-Net, we compared it with various state-of-the-art hyperspectral denoising methods, including optimization model-based approaches such as SSTPTV (2026) [52], and HLTVSG (2025) [53]; deep learning-based methods such as SERT (2023) [54], TDSAT (2024) [55], and SSRT (2024) [56]; as well as hybrid methods combining model and deep learning techniques, such as ILRNet (2024) [57]. All parameters for these algorithms were adopted from the optimal settings reported in their respective original references.

4) Evaluation Indexes: We employ PSNR, SSIM, and SAM to quantitatively evaluate the denoising performance of all methods. PSNR and SSIM, respectively, measure the reconstruction quality of lossy compression and the perceptual variations in structural information. SAM characterizes the overall spectral discrepancy between the clean and denoised HSI. A smaller SAM, along with larger PSNR and SSIM values, indicates higher image quality.

5) Experiments Setting: Real HSI may contain several distinct types of noise, including Gaussian noise, impulse noise, dead lines, and striping noise. To simulate these scenarios as comprehensively as possible, we compared six different simulated noise conditions:

1) *Case1 (Gaussian Noise + Impulse Noise):* We added zero-mean Gaussian noise with random variance and impulse noise to each band of the HSI. The variance of Gaussian noise was set to G = [0.01–0.2], and the density of the impulse noise was set to [0.01–0.1].
2) *Case2 (Gaussian Noise + Impulse Noise + Vertical stripes):* Based on Case 1, vertical stripe noise was added to 40% of the randomly selected bands.
3) *Case3 (Gaussian Noise + Impulse Noise + Horizontal stripes):* Based on Case 1, horizontal stripe noise was added to 40% of the randomly selected bands.
4) *Case4 (Gaussian Noise + Impulse Noise + Vertical stripes + Deadlines):* Based on Case 2, deadlines were added to 40% of the randomly selected bands.
5) *Case5 (Gaussian Noise + Impulse Noise + Vertical stripes + Deadlines + Wild Stripes):* Based on Case 4, wide stripe noise was added to 40% of the randomly selected bands.
6) *Case6 (Gaussian Noise + Impulse Noise + Oblique stripe + Deadlines):* Based on Case 1, tilted stripe noise and dead lines were added to 40% of the randomly selected bands.

B. Simulated Experiments

1) Quantitative Comparison: We evaluated the simulated mixed noise removal performance on the Pavia and WDC datasets across six distinct degradation scenarios, with quantitative results summarized in Table II. Under simple Gaussian and impulse noise (Case 1), both model-driven and deep learning methods perform effectively; OIC-Net yields a marginal 0.08 dB MPSNR improvement over the second-best approach (SERT). When structural stripes are introduced (Cases 2 and 3), OIC-Net consistently outperforms generic denoisers (e.g., HLTVSG, TDSAT) that lack explicit stripe modeling. While the quantitative margins are narrower when compared to dedicated destriping methods (e.g., SSTPTV), OIC-Net remains highly competitive.

The advantages of OIC-Net become highly pronounced under severe, realistic mixed-noise conditions (Cases 4 and 5). By synergistically unfolding interpretable physical models into a customized optimization network, our method transcends the inherent limitations of existing paradigms. In particular, it overcomes the inadequate representation capacity of purely

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TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON WDC AND PAVIA DATASETS IN CASES 1–6. THE BEST VALUES ARE MARKED IN BOLD, AND THE SECOND-BEST VALUES ARE UNDERLINED

Dataset	PQIs	Noisy	HLTVSG	SSTPTV	SERT	TDSAT	SSRT	ILRNet	Proposed
Pavia	Case1: Gaussian Noise + Impulse Noise								
	MPSNR	22.54	33.72	33.44	34.54	32.31	33.28	34.11	**34.62**
	MSSIM	0.581	0.942	0.941	0.945	0.921	0.938	0.946	**0.956**
	SAM	0.266	0.065	0.069	0.061	0.082	0.086	0.067	**0.057**
	Case2: Gaussian Noise + Impulse Noise + Vertical stripes								
	MPSNR	20.97	32.53	31.38	31.61	30.67	31.73	31.11	**32.88**
	MSSIM	0.487	0.934	0.918	0.932	0.911	0.914	0.924	**0.938**
	SAM	0.248	0.071	0.095	0.077	0.084	0.094	0.080	**0.075**
	Case3: Gaussian Noise + Impulse Noise + Horizontal stripes								
	MPSNR	20.75	33.42	31.00	31.97	31.46	31.44	31.04	**33.25**
	MSSIM	0.478	0.926	0.914	0.929	0.917	0.915	0.911	**0.939**
	SAM	0.247	0.097	0.107	0.084	0.081	0.099	0.101	**0.074**
	Case4: Gaussian Noise + Impulse Noise + Vertical Stripe + Deadlines								
	MPSNR	17.43	32.12	30.70	31.54	30.14	30.96	30.39	**32.32**
	MSSIM	0.461	0.918	0.910	0.903	0.901	0.906	0.899	**0.923**
	SAM	0.291	0.113	0.119	0.118	0.121	0.123	0.124	**0.112**
	Case5 Gaussian Noise + Impulse Noise + Deadlines + Wild Stripes								
	MPSNR	18.46	33.79	31.14	32.18	31.48	31.91	31.43	**33.90**
	MSSIM	0.561	0.930	0.917	0.932	0.915	0.920	0.916	**0.939**
	SAM	0.295	0.099	0.105	0.096	0.090	0.109	0.099	**0.072**
	Case6: Gaussian Noise + Impulse Noise + Oblique stripe + Deadlines								
	MPSNR	16.62	29.92	29.38	29.88	28.73	29.47	29.13	**31.78**
	MSSIM	0.376	0.887	0.876	0.897	0.868	0.878	0.872	**0.916**
	SAM	0.302	0.118	0.123	0.119	0.119	0.121	0.126	**0.115**
	Time(s)	/	163.45	16.70	8.31	4.19	11.10	6.61	**2.71**
WDC	Case1: Gaussian Noise + Impulse Noise								
	MPSNR	20.44	34.01	33.87	34.55	34.54	34.04	34.17	**35.22**
	MSSIM	0.562	0.940	0.934	0.947	0.946	0.944	0.933	**0.955**
	SAM	0.292	0.096	0.107	0.089	0.092	0.098	0.101	**0.081**
	Case2: Gaussian Noise + Impulse Noise + Vertical stripes								
	MPSNR	18.91	33.07	32.07	32.33	31.97	31.94	32.55	**33.95**
	MSSIM	0.473	0.937	0.927	0.935	0.917	0.916	0.925	**0.946**
	SAM	0.312	0.111	0.116	0.108	0.120	0.119	0.115	**0.106**
	Case3: Gaussian Noise + Impulse Noise + Horizontal stripes								
	MPSNR	19.19	33.18	32.11	32.19	31.77	31.78	31.59	**33.97**
	MSSIM	0.517	0.933	0.925	0.933	0.912	0.915	0.915	**0.944**
	SAM	0.336	0.113	0.118	0.110	0.118	0.117	0.119	**0.107**
	Case4: Gaussian Noise + Impulse Noise + Vertical Stripe + Deadlines								
	MPSNR	17.22	31.56	30.42	30.78	30.31	29.94	30.04	**32.76**
	MSSIM	0.358	0.917	0.902	0.911	0.901	0.894	0.897	**0.925**
	SAM	0.333	0.139	0.148	0.153	0.154	0.143	0.147	**0.133**
	Case5 Gaussian Noise + Impulse Noise + Deadlines + Wild Stripes								
	MPSNR	19.84	32.99	31.68	32.23	31.29	31.76	32.81	**33.74**
	MSSIM	0.611	0.937	0.920	0.931	0.915	0.917	0.928	**0.942**
	SAM	0.322	0.109	0.116	0.111	0.119	0.114	0.112	**0.105**
	Case6: Gaussian Noise + Impulse Noise + Oblique stripe + Deadlines								
	MPSNR	16.70	29.63	28.47	29.38	26.34	25.57	25.48	**31.43**
	MSSIM	0.433	0.896	0.856	0.907	0.875	0.848	0.819	**0.916**
	SAM	0.350	0.161	0.224	0.165	0.181	0.206	0.223	**0.137**
	Time(s)	/	128.25	19.09	3.19	2.68	2.28	3.30	**2.13**

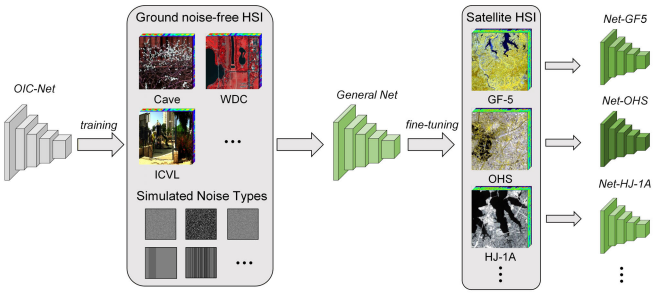


Fig. 6. Model training approach.

model-driven methods (e.g., SSTPTV) against nonlinear noise, while mitigating the poor generalization of purely data-driven approaches (e.g., SERT, TDSAT). Furthermore, compared to other model-inspired networks (e.g., ILRNet), OIC-Net achieves a deeper coupling of physical priors and features dedicated structural subnetworks, ensuring significantly better adaptation to complex real-world degradations. Finally, under the highly challenging oblique stripe scenario prevalent in UAV imagery (Case 6), OIC-Net achieves optimal quantitative metrics. This exceptional performance validates the efficacy of our learnable orientation-aware convolutional striping detection module, which precisely captures unique directional features to decouple oblique stripes from the underlying spatial textures.

2) Visual Comparison: To visually evaluate denoising performance under complex mixed noise (Case 4), Fig. 7 presents false-color images of the Pavia dataset. While comparative methods offer partial noise mitigation, they exhibit distinct drawbacks. Model-driven approaches (e.g., HLTVSG) suffer from noticeable residual stripes, spurious artifacts, and severe spectral distortion due to inadequate structural priors. Meanwhile, data-driven and hybrid methods (e.g., SSRT) generally face a rigid tradeoff between noise suppression and detail preservation, frequently resulting in either subtle residual stripes or prominent oversmoothing of high-frequency textures. On the contrary, the proposed OIC-Net thoroughly eradicates the complex noise while meticulously preserving spatial structural integrity. Furthermore, as corroborated by the spectral curves in Fig. 8, OIC-Net uniquely excels in restoring spectral fidelity, closely tracking both the global trends and local nonlinear features of the GT.

Fig. 9 illustrates the visual performance under the highly challenging oblique stripe scenario (Case 6) on the WDC dataset. Traditional model-driven methods fail to decouple these oblique patterns, as hand-crafted regularizers struggle to capture complex directional structures. Similarly, data-driven and hybrid methods either leave conspicuous structural residuals (e.g., SERT, ILRNet) or incur severe spatial blurring. Benefiting from the custom learnable orientation-aware

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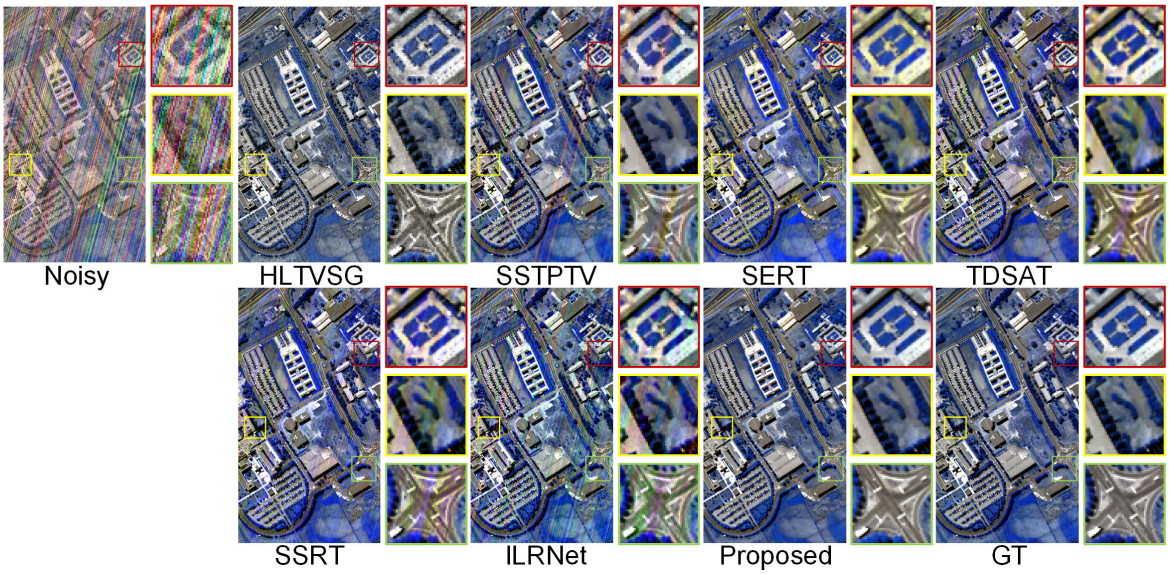


Fig. 7. Case 4 denoising results for the Pavia dataset, specifically the false-color images synthesized from bands 40, 45, and 50.

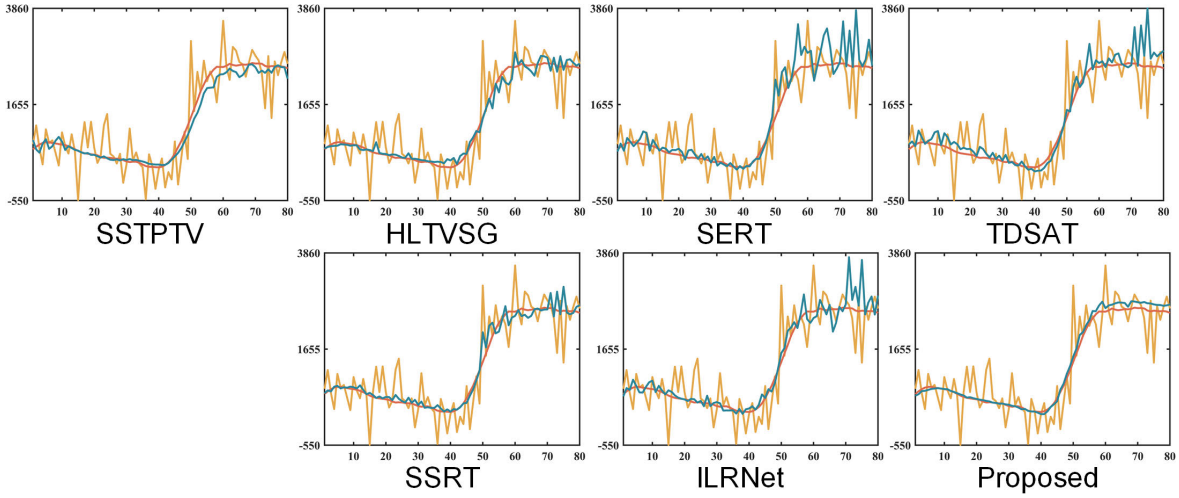


Fig. 8. Spectrum curves of the pixel (238 192) from the WDC testing dataset before and after noise removal in Case 4. Orange represents the reference spectral curve, yellow represents the degraded spectral curve, and blue represents the processed spectral curve.

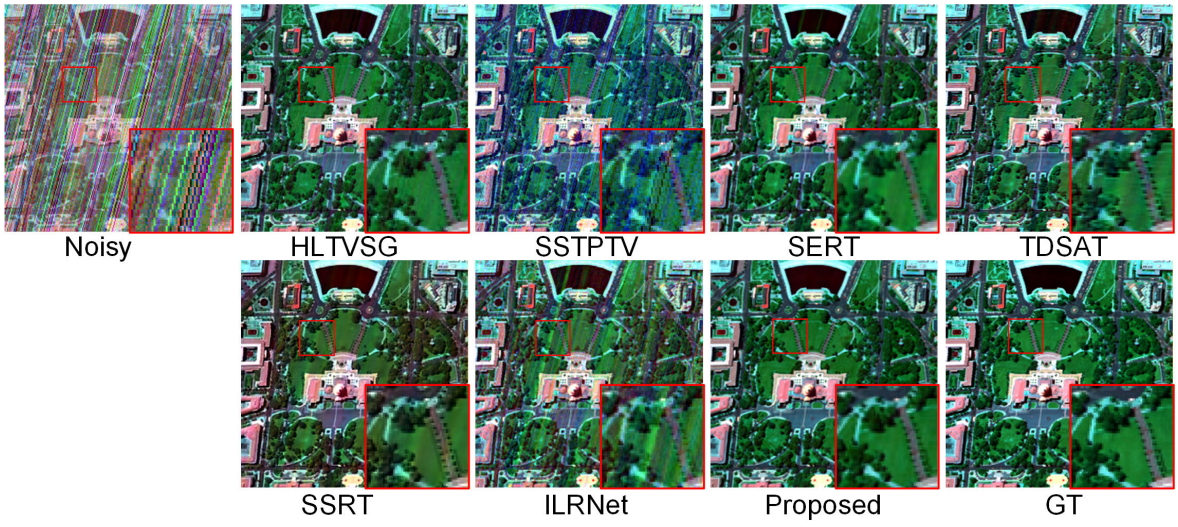


Fig. 9. Case 6 denoising results for the WDC dataset, specifically the false-color images synthesized from bands 60, 28, and 19.

convolutional striping detection module and the robust unfolded optimization framework, OIC-Net eliminates oblique artifacts while successfully reconstructing fine spatial details that are irreversibly lost by competing methods.

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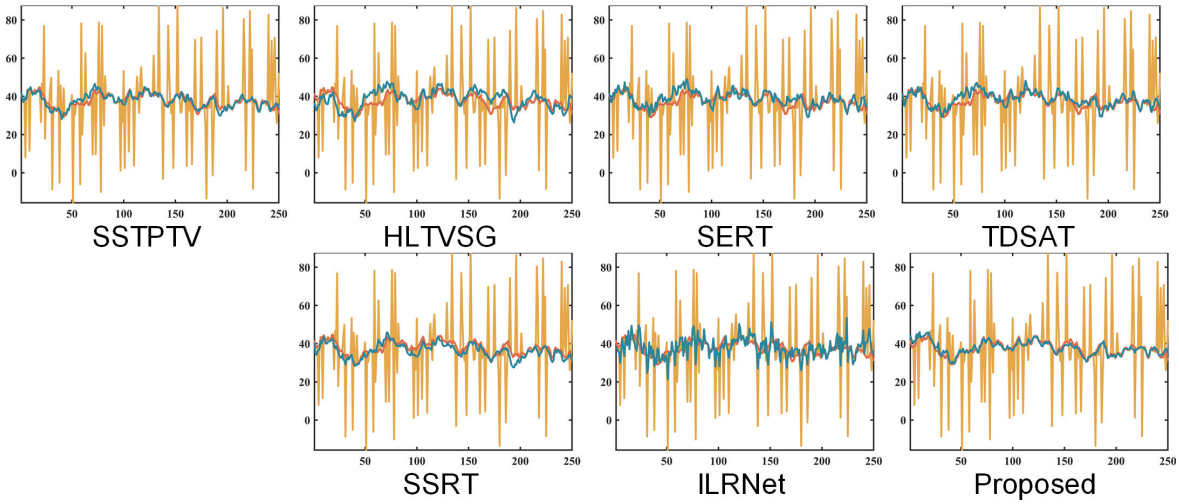


Fig. 10. Mean vertical profiles of band 106 of the WDC testing dataset before and after noise removal in Case 4. Orange represents the reference mean vertical profiles, yellow represents the degraded mean vertical profiles, and blue represents the processed mean vertical profiles.

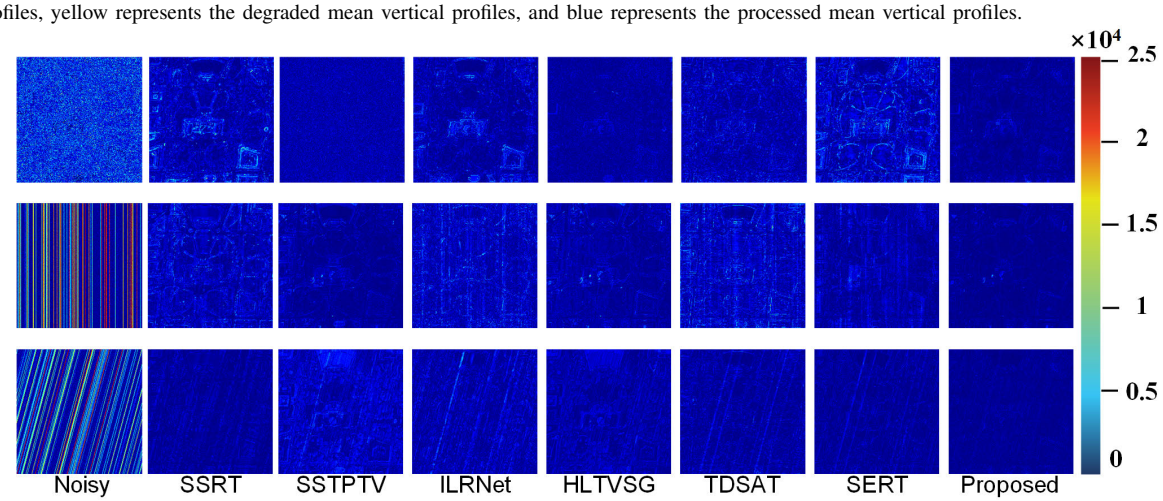


Fig. 11. From top to bottom are the error maps for the WDC synthetic noise removal results under Gaussian noise in band 110, mixed stripes in band 5, and mixed tilted stripes in band 8. These error maps illustrate the differences between the result images and the reference image.

Restoring the underlying trend of the column mean profile is a critical indicator of destriping efficacy. As shown in Fig. 10, profiles generated by comparative methods exhibit global value shifts, high-frequency jitters (HLTVSG, ILRNet), or localized fluctuations (SSRT), reflecting incomplete noise removal or spatial distortion. Conversely, the profile restored by OIC-Net exhibits the highest smoothness and strongest consistency with the GT curve. Finally, the error maps (Fig. 11) reveal that most benchmarked methods mistakenly remove underlying structural features as noise. In stark contrast, the error map of OIC-Net demonstrates the lowest overall magnitude and minimal erroneous removal of spatial details, conclusively validating its superior denoising robustness and structural preservation capability across diverse complex degradation scenarios.

C. Real Experiments

To further validate the practical effectiveness and generalization capability of the proposed OIC-Net, we evaluated its denoising performance on four real-world remote-sensing datasets corrupted by diverse and severe mixed degradations.

1) HJ-A Satellite Dataset: As illustrated in Fig. 12, while model-driven methods (e.g, HLTVSG, ILRNet) suppress the Gaussian noise, they leave noticeable residuals of the oblique stripes. Similarly, deep learning approaches (SSTPTV, SERT, TDSAT, SSRT) partially mitigate the degradation but fail to eliminate the tilted structural artifacts due to a lack of explicit directional priors. Benefiting from the unfolded iterative optimization framework and the customized learnable orientation-aware convolutional striping detection module, OIC-Net thoroughly eradicates the mixed noise while faithfully restoring underlying spatial details.

2) GF-5 Satellite Dataset: Beyond standard mixed noise, this dataset is severely afflicted by wide stripe artifacts caused by radiation distortion. As shown in Fig. 13, traditional models (HLTVSG) incur noticeable spatial blurring and stripe residuals, whereas generic deep learning approaches (SERT, TDSAT, SSRT) sacrifice fine detail preservation in heavily occluded regions. Although SSTPTV reconstructs accurate spatial structures, it introduces distinct spectral distortion (color bias). On the contrary, OIC-Net effectively eradicates the dense stripes and dead lines without compromising spectral fidelity. Driven by the interpretable numerical optimization,

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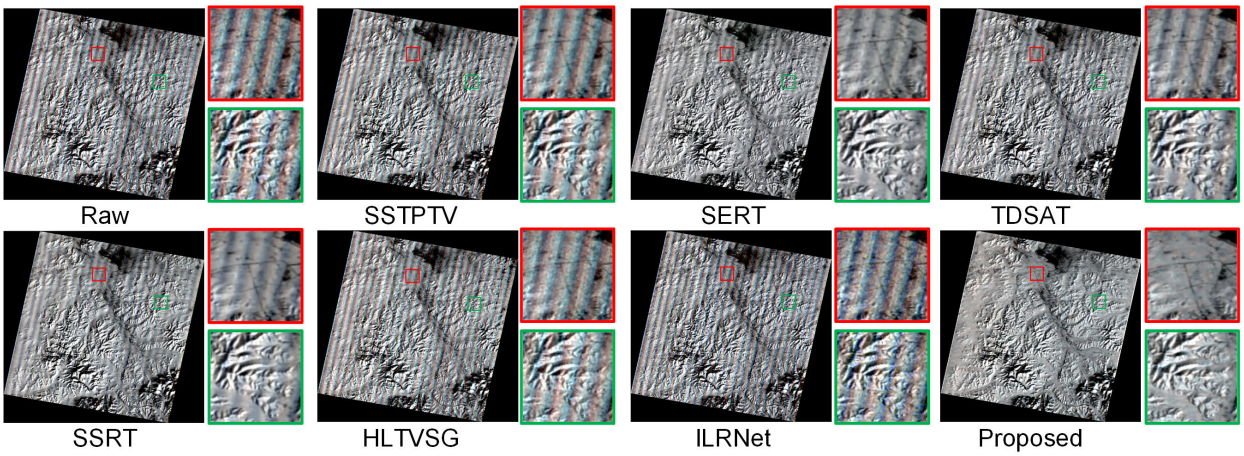


Fig. 12. Denoising results for the HJ-1A dataset, specifically the false-color image synthesized from bands 6, 7, and 8.

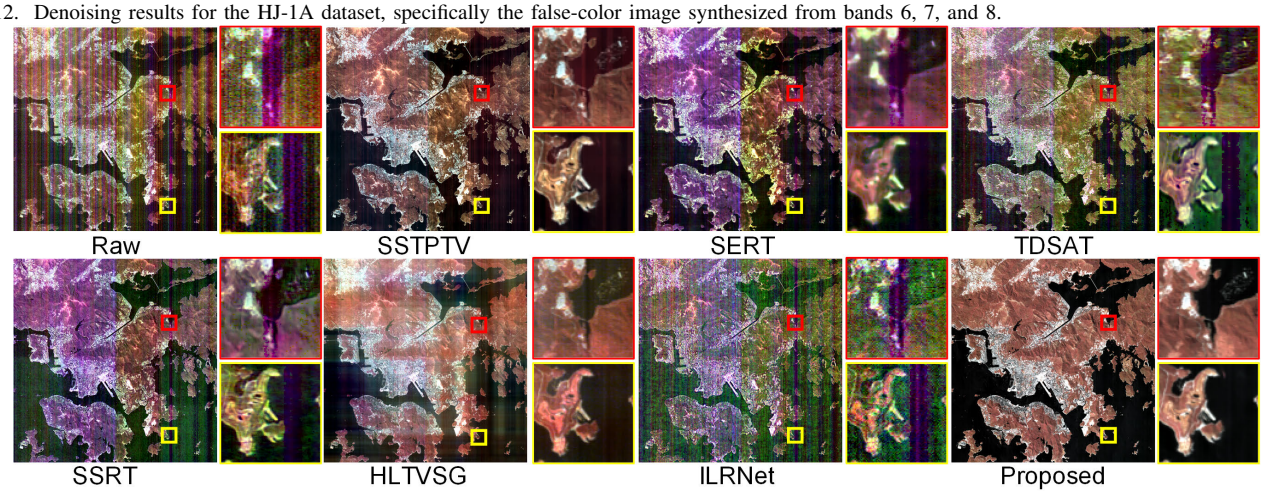


Fig. 13. Denoising results for the GF5 dataset, specifically the false-color image synthesized from bands 52, 113, and 176.

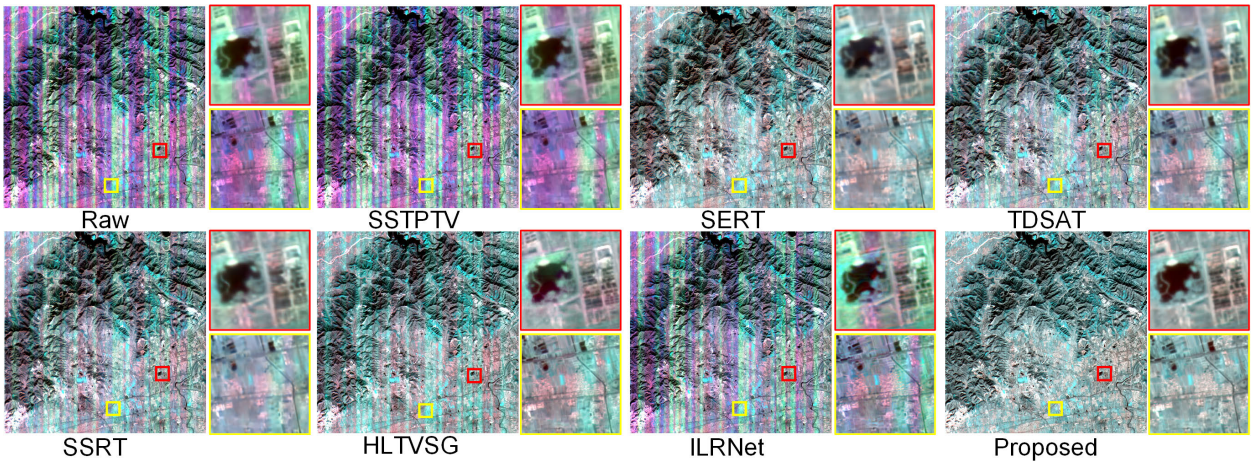


Fig. 14. Denoising results for the OHS dataset, specifically the false-color image synthesized from bands 15, 20, and 21.

our method systematically compensates for high-frequency details, yielding images with rich, faithful textures.

3) OHS Satellite Dataset: Most comparative methods struggle with the intense structural degradation in this dataset. As depicted in Fig. 14, SERT leaves conspicuous wide stripes that disrupt horizontal spatial continuity, while SSTPTV, HLTVSG, and ILRNet induce localized spectral distortions. Other data-driven models (SERT, TDSAT, SSRT) oversmooth texture details. Conversely, OIC-Net thoroughly eliminates wide stripe noise while strictly preserving structural integrity.

4) Inner Mongolia UAV Dataset: Acquired via a UAV platform, this dataset features highly challenging directional degradations accompanied by Gaussian and impulse noise

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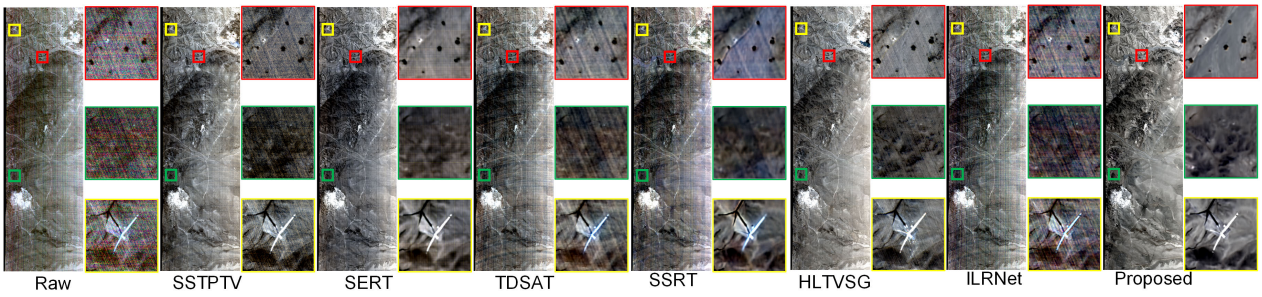


Fig. 15. Denoising results for the Inner Mongolia UAV dataset, specifically the false-color image synthesized from bands 3, 4, and 5.

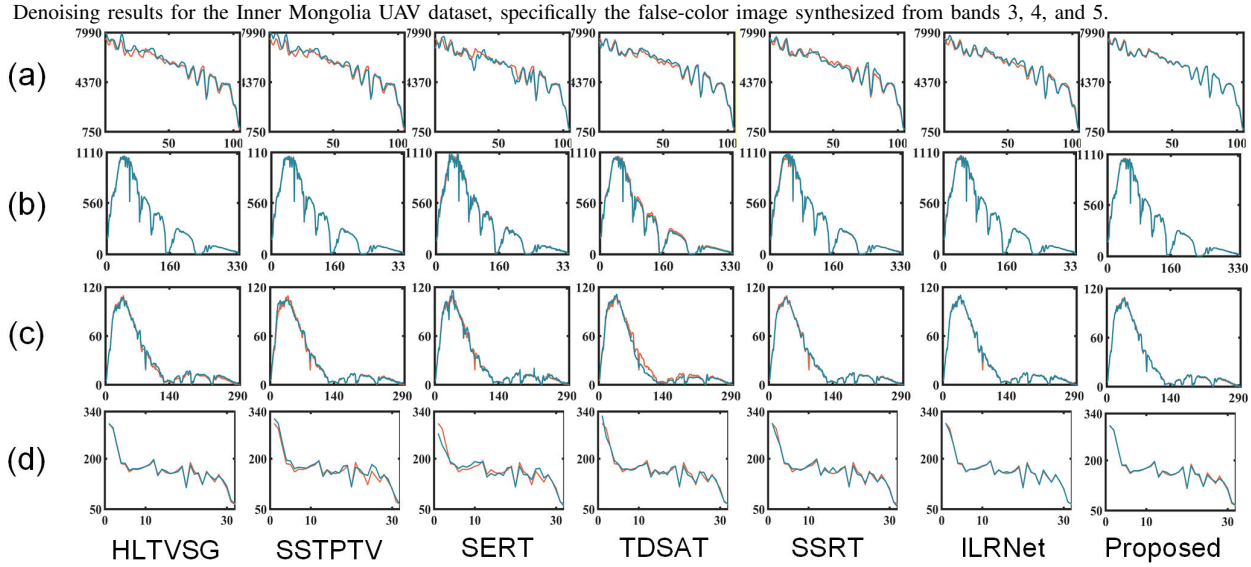


Fig. 16. Comparison of real experimental spectral curves, where orange represents the original spectral curve, and blue represents the processed spectral curve. (a) HJ-1A dataset. (b) Inner Mongolia dataset. (c) GF5 dataset. (d) OHS dataset.

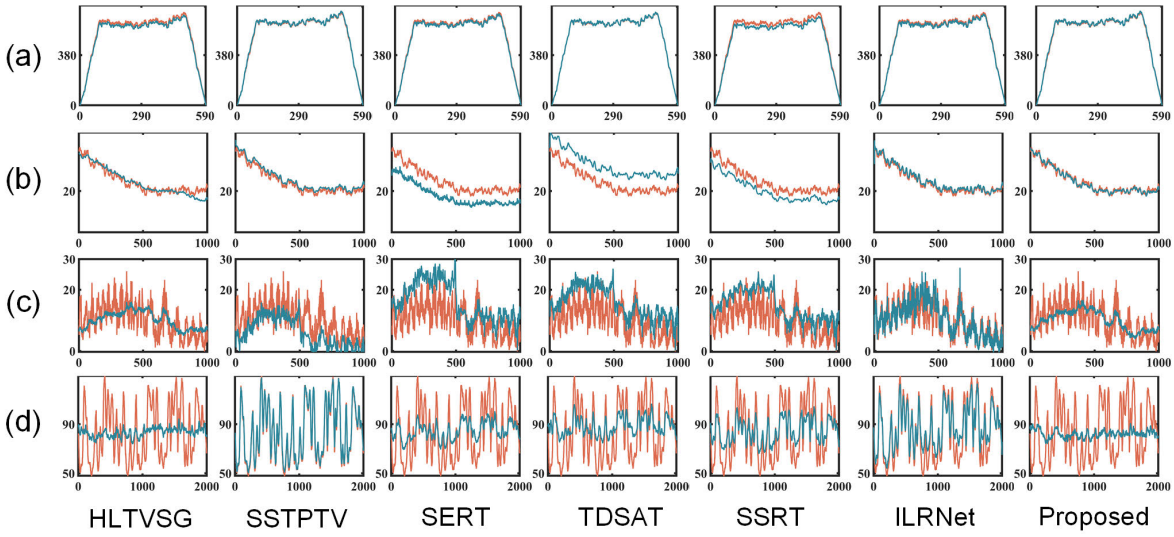


Fig. 17. Comparison of column mean curves in real experiments, and orange represents the original column mean and blue represents the processed column mean. (a) HJ-1A dataset. (b) Inner Mongolia dataset. (c) GF5 dataset. (d) OHS dataset.

TABLE III

LAND-COVER CLASSIFICATION ACCURACY ON THE REAL AVIRIS INDIAN PINES DATASET

PQIs	Noisy	HLTVSG	SSTPTV	SERT	TDSAT	SSRT	ILRNet	Proposed
OA	0.682	0.772	0.899	0.917	0.896	0.873	0.872	**0.925**
AA	0.556	0.741	0.789	0.903	0.858	0.823	0.831	**0.908**
Kappa	0.636	0.742	0.886	0.906	0.882	0.856	0.854	**0.915**

(Fig. 15). Conventional models and basic denoisers (SSTPTV, SERT, HLTVSG) fail almost entirely to decouple the oblique stripes. Likewise, the remaining deep learning approaches exhibit conspicuous directional artifacts and severe spatial oversmoothing. Ultimately, leveraging the orientation-aware detection module, OIC-Net demonstrates a robust capabil-

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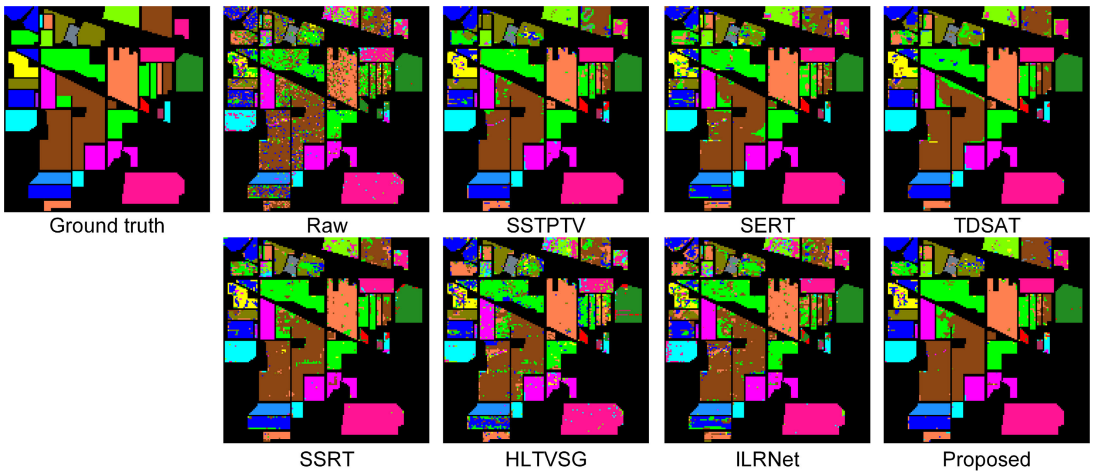


Fig. 18. Classification maps obtained from the raw and restored AVIRIS Indian Pines images using different denoising methods.

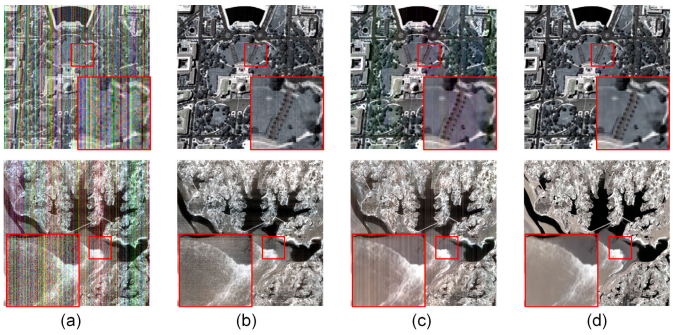


Fig. 19. Ablation study on the WDC dataset Case 4 and the GF-5 dataset. (a) Pseudo-color image contaminated by mixed noise. (b) Only the stripe detection and removal module. (c) Only the HSI denoising and reconstruction module. (d) Result of the proposed algorithm.

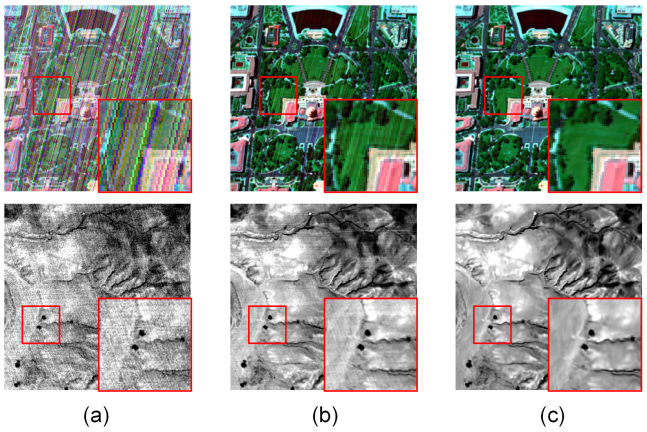


Fig. 20. Ablation studies conducted on the WDC dataset Case 6, and the drone hyperspectral dataset. (a) Pseudo-color image contaminated by mixed noise. (b) Results using only vertical and horizontal convolutions. (c) Results using all-direction convolutions.

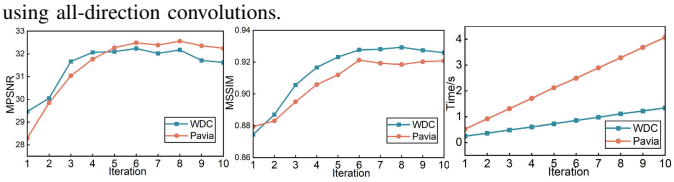


Fig. 21. Comparison of different depths of recursive unfolding on the simulated data for case 3.

ity to precisely disentangle and eliminate these complex oblique stripes, successfully reconstructing high-fidelity spatial targets. To further assess the spectral fidelity and destriping performance across the four real datasets, Figs. 16 and 17 present the spectral curves and column mean profiles, respectively. The results demonstrate that OIC-Net effectively suppresses noise-induced fluctuations while preserving the overall spectral trends and producing smoother column mean profiles.

D. Comparisons of Classification on Real Indian Pines Dataset

To quantitatively evaluate restoration performance on real HSI without clean references, we conducted a downstream classification experiment on the AVIRIS Indian Pines dataset. The support vector machine (SVM) is employed to classify all recovery results of Indian Pines under the same conditions. As shown in Table III, OIC-Net achieves the best OA, AA, and Kappa values. Compared with the noisy image, the corresponding improvements reach 0.243, 0.352, and 0.279. Moreover, Fig. 18 shows that OIC-Net generates more homogeneous regions, fewer isolated misclassifications, and clearer class boundaries. These results confirm that the proposed method effectively removes real noise while preserving spatial structures and spectral information important for downstream interpretation.

E. Discussion

1) Effectiveness of Various Parts of the Model: We evaluate the effectiveness of the MDRN module and the SPDN module. The MDRN module is used in the network to remove nonstructural noise and to refine the reconstruction of the image. The SPDN module is used to extract structural noise. From Fig. 19 and Table IV, incorporating both the MDRN and SPDN modules, can improve the network's performance in different aspects.

To confirm the effectiveness of directional convolution, we designed ablation experiments on both simulated and real

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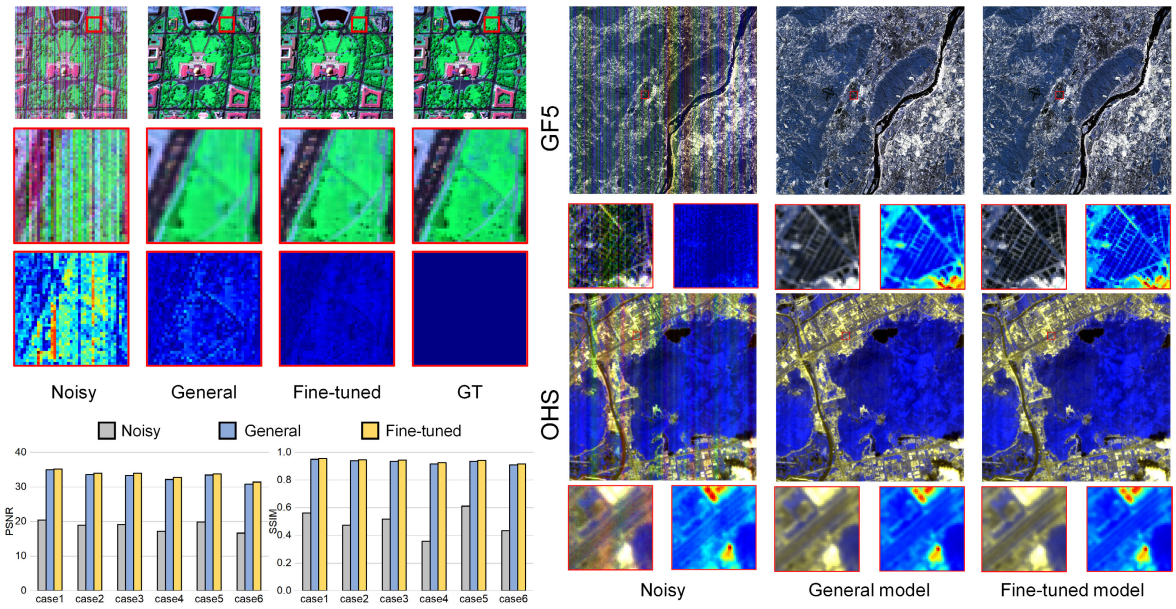


Fig. 22. Restoration results of the general model and sensor-specific fine-tuned models on different sensors.

TABLE IV

QUANTITATIVE EVALUATION OF THE FUNCTIONS OF EACH MODULE

MDRN	SPDN	MPSNR	MSSIM	SAM
√	×	31.57	0.901	0.154
×	√	31.04	0.898	0.162
√	√	**32.18**	**0.919**	**0.133**

TABLE V

QUANTITATIVE EVALUATION OF THE EFFECT OF DIRECTIONAL CONVOLUTION

PQIs	Noisy	CNN	Omnidirectional-CNN
MPSNR	16.70	29.42	**31.43**
MSSIM	0.433	0.882	**0.916**
SAM	0.350	0.198	**0.144**

TABLE VI

ABLATION STUDY OF ABUNDANCE-DOMAIN RESTORATION AND THE OPTIMIZATION-UNFOLDED ARCHITECTURE

PQIs	Noisy	HSI-domain	Cascaded	Proposed
MPSNR	17.22	31.99	31.14	**32.76**
MSSIM	0.358	0.921	0.909	**0.925**
SAM	0.333	0.138	0.150	**0.133**

datasets. Fig. 20 and Table V illustrate the performance of OIC-Net with and without directional convolution under cases containing tilted stripe noise. The results show that including directional convolution improves the visual quality and evaluation metrics of OIC-Net, validating its effectiveness.

2) Impact of Unfolded Iteration K: We explore the benefits of the recursive unfolding depth K, ranging from one to ten layers. As shown in Fig. 21, performance improves as the number of unfolded layers increases, confirming the effectiveness of the iterative network design. However, more iterations lead to a rapid growth in computational complexity, and the denoising performance tends to saturate when $K \geq 6$. By balancing performance and computational cost, we use six layers in the final model.

3) Extensibility to Different Satellite Data: Due to the scarcity of completely noise-free satellite data, it is challenging to train a model with strong generalization using a large amount of satellite data. Therefore, we train a general model using abundant noiseless ground and low-altitude drone HSI data, and then fine-tune this general model with a small amount of reference-quality satellite data to obtain a sensor-specific model. The benefit of this approach is that the fine-tuning stage preserves the noise-removal capabilities learned during the general training while adapting to the characteristics of different sensors (e.g., spatial resolution and spectral features), thereby enhancing the model's generalization and overcoming the training difficulties caused by the lack of noiseless satellite data. From Fig. 22, although the general model can effectively remove noise in satellite imagery, the fine-tuned model better reconstructs fine details that are obscured by noise.

4) Effectiveness of Abundance-Domain Restoration and the Unfolded Architecture: To further verify the necessity of the proposed model-driven design, we compared OIC-Net with two variants: direct HSI-domain restoration and a plain cascaded network. As shown in Table VI, the HSI-domain variant achieves 31.99 MPSNR, 0.921 MSSIM, and 0.138 SAM, while OIC-Net improves these results to 32.76, 0.925, and 0.133, respectively. This demonstrates that abundance-domain restoration provides a more compact and physically meaningful representation for mixed-noise removal. In addition, the cascaded variant obtains only 31.14, 0.909, and 0.150. The clear performance gap confirms that simply stacking restoration modules is insufficient, whereas the unfolded architecture benefits from explicit variable interactions and data-consistency constraints derived from the optimization model.

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TABLE VII
COMPUTATIONAL COMPLEXITY AND PROCESSING TIME OF DIFFERENT METHODS

Dataset	Size	SSTPTV	HLTVSG	SERT	TDSAT	SSRT	ILRNet	Proposed
GF5	1000×1000×295	361.17	144.04	35.83	24.22	49.11	35.11	**22.50**
HJ-1A	628×596×115	112.16	87.25	4.58	**2.93**	7.20	4.71	3.81
OHS	2000×2000×32	369.39	275.50	44.53	**30.81**	38.11	49.91	38.06
Inner Mongolia	3072×1024×335	1464.65	723.18	216.06	**152.11**	327.71	256.88	180.82
Params(#)	256×256×31	-	-	1.91	**1.09**	2.61	2.62	2.31
FLOPs(T)	256×256×31	-	-	0.45	0.93	2.93	2.06	**0.39**

5) Model Computing Costs: Table VII compares the computational complexity and processing time of different methods. For an input of $256 \times 256 \times 31$, OIC-Net contains 2.31-M parameters and requires only 0.39 TFLOPs for a complete forward inference, including MDRN, SPDN, data-consistency updates, and all unfolded stages. Its FLOPs are lower than those of SERT, TDSAT, SSRT, and ILRNet. Moreover, OIC-Net achieves the fastest processing time on GF-5 and the second-fastest performance on the other four real datasets. These results demonstrate that the proposed dual-branch unfolding architecture provides favorable computational efficiency despite incorporating two specialized restoration subnetworks.

IV. CONCLUSION

This article presents OIC-Net, an optimization-inspired deep network designed to tackle complex mixed noise and oblique stripes in HSI. By conducting restoration in the abundance domain, the framework exploits inherent spectral redundancy and achieves band-agnostic flexibility. OIC-Net synergistically couples an abundance restoration subnetwork with a stripe removal subnetwork equipped with an omni-directional convolutional module, which explicitly models the structural priors of multidirectional stripes. Through iterative optimization-guided interactions, the network progressively suppresses severe degradations while preserving spatial-spectral fidelity. Comprehensive evaluations on synthetic and real-world datasets confirm that OIC-Net achieves state-of-the-art denoising performance, outperforming conventional and deep learning methods in generalization, oblique stripe removal, and computational efficiency. Nevertheless, completely missing bands remain difficult to reconstruct because of the absence of valid observations. Moreover, sensors with substantially different spatial resolutions may still require limited fine-tuning for optimal performance. Future work will investigate external spectral priors for missing-band reconstruction and more adaptive cross-sensor learning strategies.

REFERENCES

[1] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012, doi: 10.1109/JSTARS.2012.2194696.

[2] B. Aiazzi, L. Alparone, A. Barducci, S. Baronti, and I. Pippi, "Information-theoretic assessment of sampled hyperspectral imagers," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 7, pp. 1447–1458, Jul. 2001, doi: 10.1109/36.934076.

[3] R. Näsi et al., "Using UAV-based photogrammetry and hyperspectral imaging for mapping bark beetle damage at tree-level," *Remote Sens.*, vol. 7, no. 11, pp. 15467–15493, Nov. 2015, doi: 10.3390/rs71115467.

[4] H. Zhai, H. Zhang, P. Li, and L. Zhang, "Hyperspectral image clustering: Current achievements and future lines," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 35–67, Dec. 2021, doi: 10.1109/MGRS.2020.3032575.

[5] A. Plaza et al., "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Sep. 2009, doi: 10.1016/j.rse.2007.07.028.

[6] J. Li, H. Shen, H. Li, M. Jiang, and Q. Yuan, "Radiometric quality improvement of hyperspectral remote sensing images: A technical tutorial on variational framework," *J. Appl. Remote Sens.*, vol. 15, no. 3, Sep. 2021, Art. no. 031502, doi: 10.1117/1.jrs.15.031502.

[7] H. Shen, M. Jiang, J. Li, C. Zhou, Q. Yuan, and L. Zhang, "Coupling model- and data-driven methods for remote sensing image restoration and fusion: Improving physical interpretability," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 231–249, Jun. 2022, doi: 10.1109/MGRS.2021.3135954.

[8] X. Lu, Y. Wang, and Y. Yuan, "Graph-regularized low-rank representation for destriping of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 4009–4018, Jul. 2013, doi: 10.1109/TGRS.2012.2226730.

[9] H. Othman and S.-E. Qian, "Noise reduction of hyperspectral imagery using hybrid spatial–spectral derivative-domain wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 2, pp. 397–408, Feb. 2006, doi: 10.1109/TGRS.2005.860982.

[10] M. Elad and M. Aharon, "Image denoising via learned dictionaries and sparse representation," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2006, pp. 895–900, doi: 10.1109/CVPR.2006.142.

[11] H. Zhang, W. He, L. Zhang, H. Shen, and Q. Yuan, "Hyperspectral image restoration using low-rank matrix recovery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4729–4743, Aug. 2014, doi: 10.1109/TGRS.2013.2284280.

[12] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015, doi: 10.1109/JSTARS.2015.2398433.

[13] J. Xue, Y. Zhao, W. Liao, and J. C.-W. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019, doi: 10.1109/TGRS.2019.2897316.

[14] L. Sun and C. He, "Hyperspectral image mixed denoising using difference continuity-regularized nonlocal tensor subspace low-rank learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3090178.

[15] D. Li, D. Chu, X. Guan, W. He, and H. Shen, "Adaptive regularized low-rank tensor decomposition for hyperspectral image denoising and destriping," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–17, 2024, doi: 10.1109/TGRS.2024.3385536.

[16] J. Lin, T.-Z. Huang, X.-L. Zhao, T.-H. Ma, T.-X. Jiang, and Y.-B. Zheng, "A novel non-convex low-rank tensor approximation model for hyperspectral image restoration," *Appl. Math. Comput.*, vol. 408, Nov. 2021, Art. no. 126342, doi: 10.1016/j.amc.2021.126342.

[17] Y. Chen, J. Zhang, J. Zeng, W. Lai, X. Gui, and T.-X. Jiang, "A guidable nonlocal low-rank approximation model for hyperspectral image denoising," *Signal Processing*, vol. 215, Feb. 2024, Art. no. 109266, doi: 10.1016/j.sigpro.2023.109266.

[18] Y. Chen, W. He, N. Yokoya, T.-Z. Huang, and X.-L. Zhao, "Nonlocal tensor-ring decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1348–1362, Feb. 2020, doi: 10.1109/TGRS.2019.2946050.

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[19] H. Wang, J. Peng, W. Qin, J. Wang, and D. Meng, "Guaranteed tensor recovery fused low-rankness and smoothness," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10990–11007, Sep. 2023, doi: 10.1109/TPAMI.2023.3259640.

[20] Y. Chen, W. Cao, L. Pang, J. Peng, and X. Cao, "Hyperspectral image denoising via texture-preserved total variation regularizer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5516114, doi: 10.1109/TGRS.2023.3292518.

[21] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, Y. Chen, and W. He, "Double-factor-regularized low-rank tensor factorization for mixed noise removal in hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8450–8464, Dec. 2020, doi: 10.1109/TGRS.2020.2987954.

[22] X. Liu, X. Lu, H. Shen, Q. Yuan, Y. Jiao, and L. Zhang, "Stripe noise separation and removal in remote sensing images by consideration of the global sparsity and local variational properties," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 3049–3060, May 2016, doi: 10.1109/TGRS.2015.2510418.

[23] H. Zhang, J. Cai, W. He, H. Shen, and L. Zhang, "Double low-rank matrix decomposition for hyperspectral image denoising and destriping," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502619, doi: 10.1109/TGRS.2021.3061148.

[24] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.

[25] W. Xie, Y. Li, and X. Jia, "Deep convolutional networks with residual learning for accurate spectral–spatial denoising," *Neurocomputing*, vol. 312, pp. 372–381, Oct. 2018, doi: 10.1016/j.neucom.2018.05.115.

[26] W. Dong, H. Wang, F. Wu, G. Shi, and X. Li, "Deep spatial–spectral representation learning for hyperspectral image denoising," *IEEE Trans. Comput. Imag.*, vol. 5, no. 4, pp. 635–648, Dec. 2019.

[27] K. Wei, Y. Fu, and H. Huang, "3-D quasi-recurrent neural network for hyperspectral image denoising," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 363–375, Jan. 2021.

[28] E. Pan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "SQAD: Spatial–spectral quasi-attention recurrent network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524814.

[29] H. Zeng et al., "Degradation-noise-aware deep unfolding transformer for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–12, 2025, doi: 10.1109/TGRS.2025.3543920.

[30] C. Chen, J. Li, X. Liu, Q. Yuan, and L. Zhang, "Bidirectional-aware network combining transformer and mamba for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5514316, doi: 10.1109/TGRS.2025.3565613.

[31] M. Li, Y. Fu, and Y. Zhang, "Spatial–spectral transformer for hyperspectral image denoising," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 1, pp. 1368–1376.

[32] Z. Wu, Y. Chen, N. Yokoya, and W. He, "MP-HSIR: A multi-prompt framework for universal hyperspectral image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 13009–13020.

[33] A. Dixit, A. K. Gupta, P. Gupta, and A. Garg, "RECREATE: Supervised contrastive learning and inpainting based hyperspectral image denoising," *ISPRS J. Photogramm. Remote Sens.*, vol. 233, pp. 14–24, Mar. 2026, doi: 10.1016/j.isprsjprs.2026.01.022.

[34] J. Peng et al., "Learnable representative coefficient image denoiser for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506516.

[35] F. Xiong, J. Zhou, J. Zhou, J. Lu, and Y. Qian, "Deep parameterized neural networks for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525015, doi: 10.1109/TGRS.2023.3318001.

[36] Y. Chang, L. Yan, T. Wu, and S. Zhong, "Remote sensing image stripe noise removal: From image decomposition perspective," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7018–7031, Dec. 2016, doi: 10.1109/TGRS.2016.2594080.

[37] Y. Li, J. Li, J. He, X. Liu, and Q. Yuan, "An optimization-driven network with knowledge prior injection for HSI denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531717.

[38] Y. Chang, L. Yan, H. Fang, and C. Luo, "Anisotropic spectral–spatial total variation model for multispectral remote sensing image destriping," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1852–1866, Jun. 2015, doi: 10.1109/TIP.2015.2404782.

[39] B. Li et al., "Synthesis and detection algorithms for oblique stripe noise of space-borne remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024, doi: 10.1109/TGRS.2024.3360268.

[40] J.-L. Wang, T.-Z. Huang, T.-H. Ma, X.-L. Zhao, and Y. Chen, "A sheared low-rank model for oblique stripe removal," *Appl. Math. Comput.*, vol. 360, pp. 167–180, Nov. 2019, doi: 10.1016/j.amc.2019.03.066.

[41] H. Wu, X. Jiang, Y. Li, L. Gui, and F. Xiao, "Robust deep low-rank tensor factorization for hyperspectral image restoration," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Apr. 29, 2026, doi: 10.1109/TCSVT.2026.3688550.

[42] L. Zhuang, M. K. Ng, L. Gao, J. Michalski, and Z. Wang, "Eigenimage2Eigenimage (E2E): A self-supervised deep learning network for hyperspectral image denoising," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16262–16276, Nov. 2024, doi: 10.1109/TNNLS.2023.3293328.

[43] J. Ye, J. Wang, F. Xiong, J. Chen, and Y. Qian, "Deep equilibrium convolutional sparse coding for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5523014, doi: 10.1109/TGRS.2025.3603390.

[44] Y. Li, X. Jiang, L. Gui, and F. Xiao, "Robust deep recovery model with spatial–spectral total generalized variation prior for hyperspectral image denoising," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 4, pp. 4091–4105, Apr. 2026, doi: 10.1109/TCSVT.2025.3637370.

[45] W. He, Q. Yao, C. Li, N. Yokoya, and Q. Zhao, "Non-local meets global: An integrated paradigm for hyperspectral denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 6861–6870, doi: 10.1109/CVPR.2019.00703.

[46] Y.-C. Miao, X.-L. Zhao, X. Fu, J.-L. Wang, and Y.-B. Zheng, "Hyperspectral denoising using unsupervised disentangled spatiospectral deep priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513916, doi: 10.1109/TGRS.2021.3106380.

[47] Y. Miao, L. Zhang, L. Zhang, and D. Tao, "DDS2M: Self-supervised denoising diffusion spatio-spectral model for hyperspectral image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 12052–12062, doi: 10.1109/ICCV51070.2023.01110.

[48] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005, doi: 10.1109/TGRS.2005.844293.

[49] R. Alhejaili, M. Alfarraj, H. Luqman, and A. Al-Shaikhi, "Recursions are all you need: Towards efficient deep unfolding networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 4705–4714, doi: 10.1109/CVPRW59228.2023.00498.

[50] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481, doi: 10.1109/CVPR.2018.00262.

[51] E. Pan, Y. Ma, X. Mei, J. Huang, Q. Chen, and J. Ma, "Hyperspectral image destriping and denoising from a task decomposition view," *Pattern Recognit.*, vol. 144, Dec. 2023, Art. no. 109832, doi: 10.1016/j.patcog.2023.109832.

[52] J.-L. Zhao, T.-H. Zhang, S. Fang, J.-F. Gao, J.-Y. Wang, and M.-G. Gong, "Spatial–spectral texture-preserved total variation: A novel regularization for hyperspectral image denoising," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 1, pp. 248–260, Jan. 2026, doi: 10.1109/TCSVT.2025.3598368.

[53] F. Yang, Q. Hu, and X. Su, "Hyperspectral image denoising based on hyper-laplacian total variation in spectral gradient domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–17, 2025, doi: 10.1109/TGRS.2025.3549526.

[54] M. Li, J. Liu, Y. Fu, Y. Zhang, and D. Dou, "Spectral enhanced rectangle transformer for hyperspectral image denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5805–5814, doi: 10.1109/CVPR52729.2023.00562.

[55] Q. Zhang et al., "Three-dimension spatial–spectral attention transformer for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024, doi: 10.1109/TGRS.2024.3458174.

[56] G. Fu, F. Xiong, J. Lu, J. Zhou, J. Zhou, and Y. Qian, "Hyperspectral image denoising via spatial–spectral recurrent transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511214, doi: 10.1109/TGRS.2024.3374953.

[57] J. Ye, F. Xiong, J. Zhou, and Y. Qian, "Iterative low-rank network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528015, doi: 10.1109/TGRS.2024.3449130.

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